

PROJECT REPORT ON

Predicting Osteoporosis Risk

DONE BY:

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1. Project Introduction

Osteoporosis is a systemic skeletal condition marked by decreased bone mineral density (BMD) and micro-architectural deterioration of bone tissue. This deterioration leads to heightened skeletal fragility and a substantially elevated risk of low-trauma fractures. Because osteoporosis progresses silently without manifest symptoms until a fracture occurs, early screening is essential.

Aim & Objectives

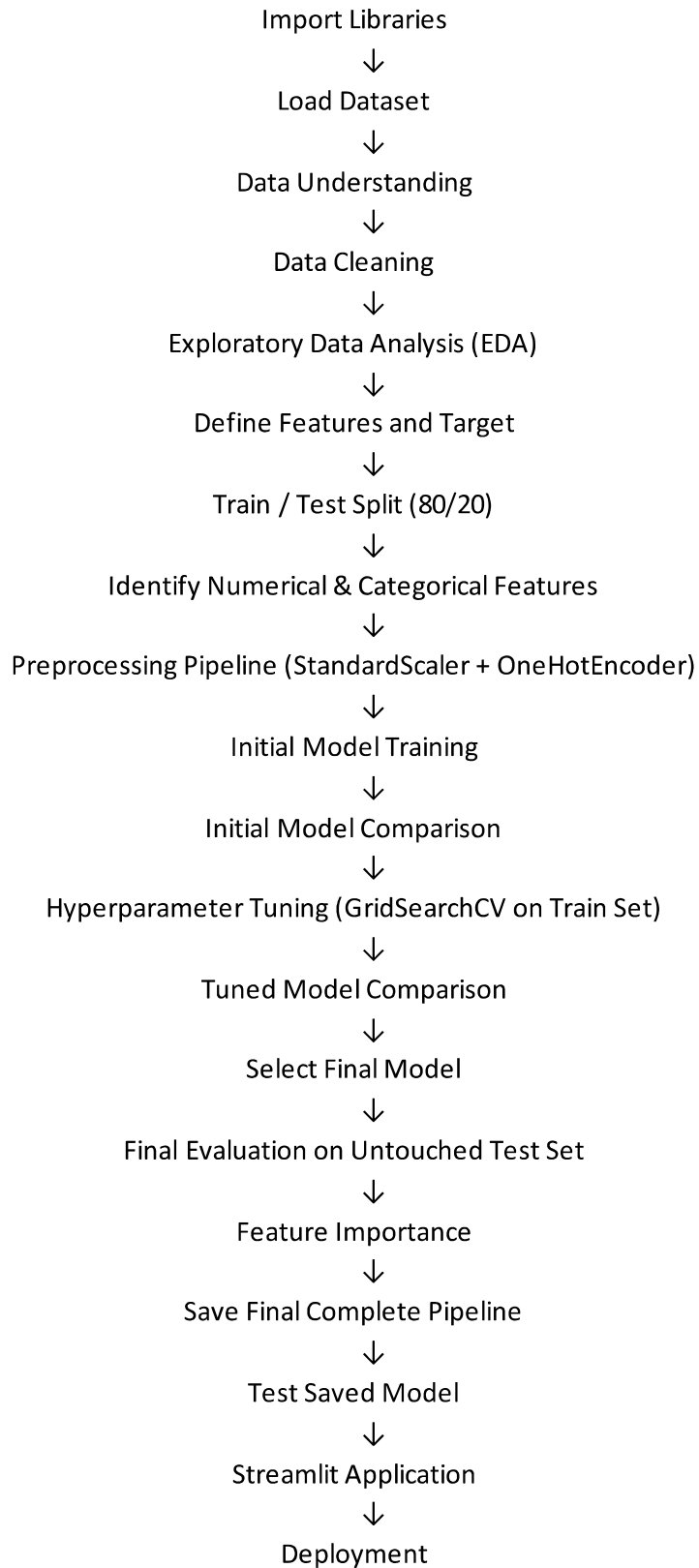
- Aim: Develop a machine learning classification model to predict individual osteoporosis risk.
- Objective: Facilitate early diagnostic intervention to optimize patient care and reduce fragility fracture incidence.

2. Dataset Summary

- Source: osteoporosis.csv
- Dimensions: 1,958 rows * 16 columns
- Target Variable: Osteoporosis (Binary: 0 = Negative, 1 = Positive; {exactly 50% / 50% balanced})
- Features:
 - 1 Continuous Numerical: Age
 - 14 Categorical: Gender, Hormonal Changes, Family History, Race/Ethnicity, Body Weight, Calcium Intake, Vitamin D Intake, Physical Activity, Smoking, Alcohol Consumption, Medical Conditions, Medications, Prior Fractures
 - 1 Identifier: Id (dropped)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Id	Age	Gender	Hormonal Changes	Family History	Race/Ethnicity	Body Weight	Calcium Intake	Vitamin D Intake	Physical Activity	Smoking	Alcohol Consumption	Medical Conditions	Medications	Prior Fractures	Osteoporosis
2	104866	69	Female	Normal	Yes	Asian	Underweight	Low	Sufficient	Sedentary	Yes	Moderate	Rheumatoid Arthritis	Corticosteroids	Yes	1
3	101999	32	Female	Normal	Yes	Asian	Underweight	Low	Sufficient	Sedentary	No	None	None	None	Yes	1
4	106567	89	Female	Postmenopausal	No	Caucasian	Normal	Adequate	Sufficient	Active	No	Moderate	Hyperthyroidism	Corticosteroids	No	1
5	102316	78	Female	Normal	No	Caucasian	Underweight	Adequate	Insufficient	Sedentary	Yes	None	Rheumatoid Arthritis	Corticosteroids	No	1
6	101944	38	Male	Postmenopausal	Yes	African American	Normal	Low	Sufficient	Active	Yes	None	Rheumatoid Arthritis	None	Yes	1
7	102265	41	Male	Normal	Yes	Caucasian	Normal	Low	Sufficient	Active	Yes	Moderate	Rheumatoid Arthritis	Corticosteroids	Yes	1
8	107447	20	Male	Postmenopausal	Yes	African American	Underweight	Adequate	Sufficient	Sedentary	No	None	Rheumatoid Arthritis	None	No	1
9	103065	39	Male	Postmenopausal	Yes	Asian	Normal	Adequate	Sufficient	Sedentary	No	None	Rheumatoid Arthritis	Corticosteroids	Yes	1
10	103040	70	Male	Postmenopausal	No	Asian	Underweight	Low	Sufficient	Active	Yes	None	Rheumatoid Arthritis	Corticosteroids	No	1
11	105960	19	Female	Normal	No	African American	Normal	Low	Sufficient	Active	Yes	Moderate	None	Corticosteroids	Yes	1
12	101592	47	Female	Postmenopausal	Yes	Asian	Normal	Low	Sufficient	Active	Yes	None	None	None	Yes	1
13	103872	55	Female	Normal	Yes	Caucasian	Underweight	Adequate	Sufficient	Sedentary	No	Moderate	Rheumatoid Arthritis	Corticosteroids	No	1
14	102439	19	Female	Postmenopausal	Yes	Asian	Underweight	Low	Insufficient	Active	Yes	None	None	Corticosteroids	Yes	1
15	104084	81	Male	Normal	Yes	Caucasian	Underweight	Adequate	Insufficient	Sedentary	Yes	Moderate	Hyperthyroidism	Corticosteroids	No	1
16	105371	77	Male	Normal	Yes	African American	Underweight	Low	Sufficient	Sedentary	Yes	None	Hyperthyroidism	None	No	1
17	109617	38	Male	Postmenopausal	Yes	African American	Normal	Adequate	Sufficient	Active	Yes	None	Rheumatoid Arthritis	None	No	1
18	109183	50	Female	Postmenopausal	No	Asian	Underweight	Adequate	Sufficient	Active	Yes	Moderate	Hyperthyroidism	None	No	1
19	106854	75	Male	Postmenopausal	No	Asian	Normal	Adequate	Sufficient	Sedentary	No	None	Hyperthyroidism	None	No	1
20	104601	39	Female	Postmenopausal	No	Caucasian	Normal	Adequate	Insufficient	Sedentary	No	Moderate	None	None	No	1
21	103988	66	Male	Postmenopausal	Yes	Caucasian	Normal	Low	Insufficient	Sedentary	No	None	None	None	Yes	1
22	107007	76	Male	Normal	Yes	Asian	Normal	Adequate	Sufficient	Sedentary	Yes	Moderate	None	Corticosteroids	No	1
23	105868	59	Female	Postmenopausal	Yes	Asian	Normal	Adequate	Insufficient	Active	No	None	None	Corticosteroids	No	1

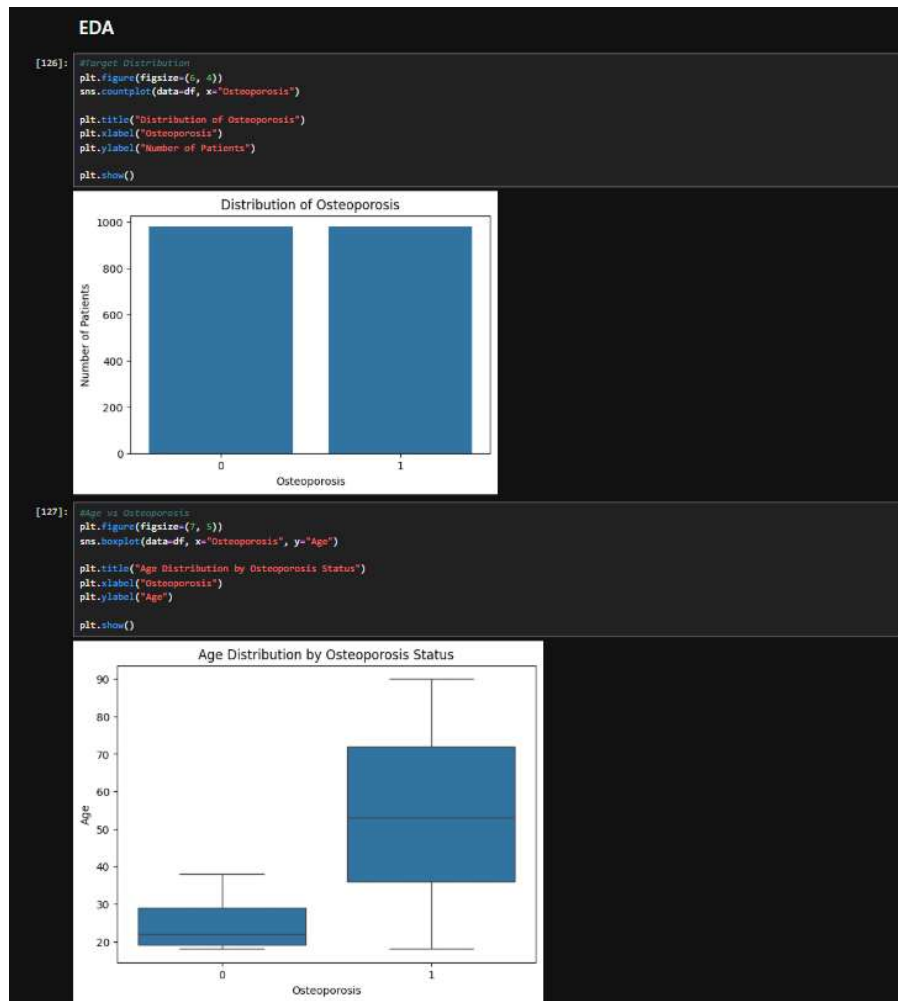
3. End-to-End Workflow



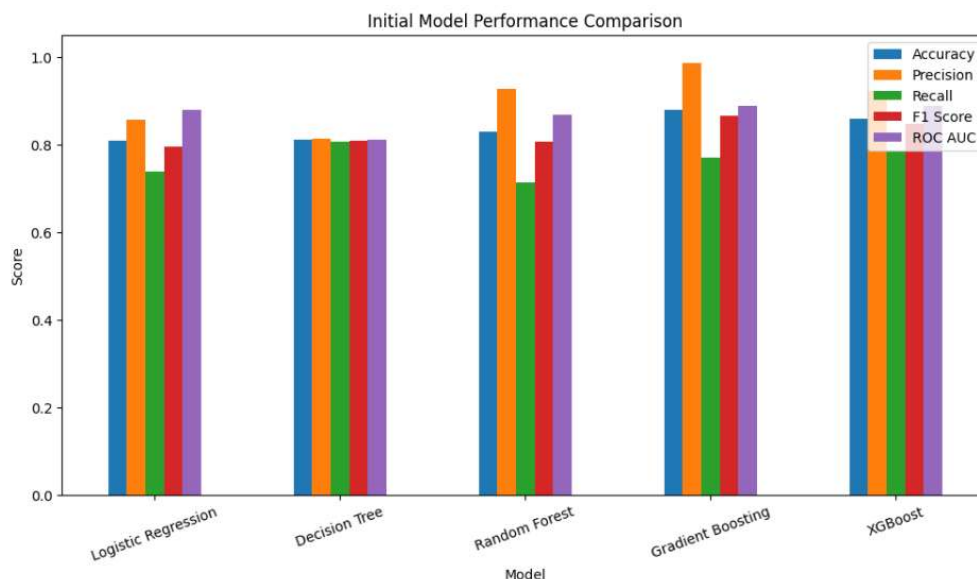
4. Notebook Implementation

- **Data Cleaning:** Confirmed 0 duplicate rows. Imputed missing values in Alcohol Consumption (988), Medications (985), and Medical Conditions (647) with "Not Reported". Dropped the uninformative Id column.

- **EDA:** Target distribution is 979 cases each. Box plots indicate strong positive separation by Age (non-osteoporotic cases concentrate around age 24, positive cases skew significantly older up to 90). High risk aligns with postmenopausal status, prior fractures, low activity, and underweight status.



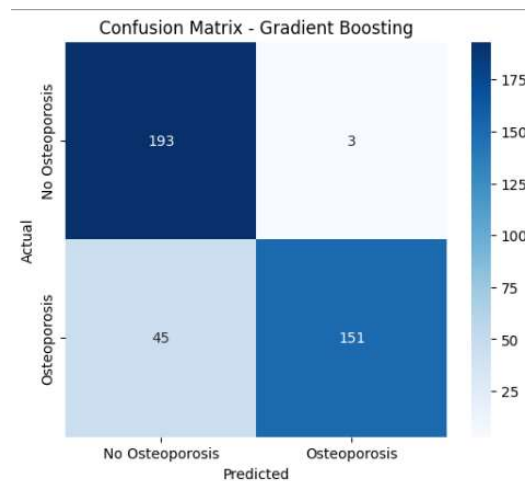
- **Feature Processing & Split:** 80/20 stratified split (1,566 train / 392 test). Bundled into a ColumnTransformer (StandardScaler on Age, OneHotEncoder(handle_unknown='ignore') on all 13 categorical features).
- **Initial Models Tested:** Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost. Gradient Boosting produced the top initial baseline (F₁ = 0.8653).



- **Hyperparameter Tuning:** Ran 5-fold cross-validation grid search on the training split. Gradient Boosting parameters settled at `n_estimators=100`, `learning_rate=0.05`, `max_depth=4`, `subsample=0.8` (CV F1 = 0.9141).

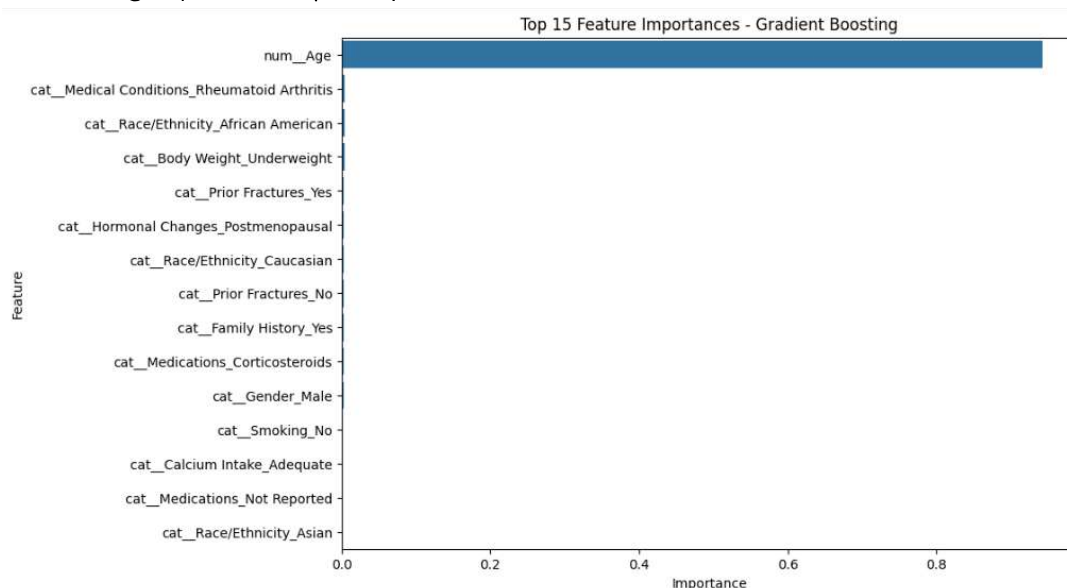
5. Final Model Evaluation & Scores

- **Selected Model:** Tuned Gradient Boosting Classifier
- **Test Set Performance (N = 392):**
 - Accuracy: 87.76% (~0.88)
 - Precision: 98.05% (~0.98)
 - Recall: 77.04% (~0.77)
 - F1-Score: 86.29% (~0.86)
 - ROC-AUC: 88.71% (~0.89)
- **Confusion Matrix:**



6. Top Feature Importances

1. Age: 94.23%
2. Medical Conditions (Rheumatoid Arthritis): 0.42%
3. Race/Ethnicity (African American): 0.40%
4. Body Weight (Underweight): 0.36%
5. Prior Fractures (Yes): 0.32%
6. Hormonal Changes (Postmenopausal): 0.32%



7. Streamlit Application & Deployment

- File: app_osteoporosis.py
- Features: Interactive input forms for all 14 clinical attributes, live predicted risk probability gauge, low/high risk status badge, and dynamic clinical guidance (DEXA scan referrals, Vitamin D3/Calcium intake, medication review).
- Local Run: streamlit run app_osteoporosis.py via Anaconda Prompt.
- Production Deployment: Hosted via Streamlit Community Cloud linked to the project GitHub repository.

This screenshot shows the initial state of the OsteoGuard AI application. The interface is divided into two main sections: input forms on the left and clinical recommendations on the right.

Input Forms:

- Race / Ethnicity:** Caucasian
- Family History of Osteoporosis:** Yes
- Nutrition & Daily Lifestyle:**
 - Dietary Calcium Intake:** Low
 - Smoking Status:** Yes
 - Vitamin D Intake:** Insufficient
 - Alcohol Consumption:** Moderate
 - Physical Activity Level:** Sedentary
- Medical History & Pharmacotherapy:**
 - Underlying Secondary Medical Conditions:** Rheumatoid Arthritis
 - History of Prior Low-Trauma Fractures:** Yes
 - Prescribed High-Risk Medications:** Corticosteroids

Clinical Recommendations:

- Order a diagnostic Dual-Energy X-ray Absorptiometry (DEXA) scan of lumbar spine and femoral neck.
- Recommend dietary calcium optimization (1,200 mg/day elemental calcium equivalent).
- Prescribe cholecalciferol (Vitamin D3) supplementation (800-2,000 IU daily).
- Encourage progressive weight-bearing aerobic and muscle-strengthening exercise programs.
- Review long-term corticosteroid therapy regimen; evaluate concurrent prophylactic bisphosphonates.
- History of prior fracture elevates subsequent fracture hazard; consider initiating antiresorptive or bone-anabolic therapy.

Footer: Disclaimer: OsteoGuard AI is an adjunctive decision support tool. All clinical decisions must be guided by verified BMD testing, FRAX scoring, and board-certified physician discretion.

This screenshot shows the OsteoGuard AI application with a high-risk assessment. The input forms are filled out, and the clinical risk assessment output is displayed.

Patient Clinical & Lifestyle Profile:

- Demographics & Constitutional Factors:**
 - Patient Age (years):** 72
 - Body Weight Category:** Underweight
 - Biological Sex:** Female
 - Hormonal Profile:** Postmenopausal
 - Race / Ethnicity:** Caucasian
 - Family History of Osteoporosis:** Yes
- Nutrition & Daily Lifestyle:**
 - Dietary Calcium Intake:** Low
 - Smoking Status:** Yes
 - Vitamin D Intake:** Insufficient
 - Alcohol Consumption:** Moderate
 - Physical Activity Level:** Sedentary

Clinical Risk Assessment Output:

Predicted Osteoporosis Probability: 99.3%

High Risk Detected (Score: 99.3%)

The patient exhibits high-risk indicators associated with secondary and primary bone demineralization.

Clinical Recommendations:

- Order a diagnostic Dual-Energy X-ray Absorptiometry (DEXA) scan of lumbar spine and femoral neck.
- Recommend dietary calcium optimization (1,200 mg/day elemental calcium equivalent).
- Prescribe cholecalciferol (Vitamin D3) supplementation (800-2,000 IU daily).
- Encourage progressive weight-bearing aerobic and muscle-strengthening exercise programs.
- Review long-term corticosteroid therapy regimen; evaluate concurrent prophylactic bisphosphonates.
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This screenshot shows the OsteoGuard AI application with a low-risk assessment. The input forms are filled out, and the clinical risk assessment output is displayed.

Patient Clinical & Lifestyle Profile:

- Demographics & Constitutional Factors:**
 - Patient Age (years):** 72
 - Body Weight Category:** Normal
 - Biological Sex:** Female
 - Hormonal Profile:** Normal
 - Race / Ethnicity:** Caucasian
 - Family History of Osteoporosis:** No
- Nutrition & Daily Lifestyle:**
 - Dietary Calcium Intake:** Adequate
 - Smoking Status:** No
 - Vitamin D Intake:** Sufficient
 - Alcohol Consumption:** Not Reported
 - Physical Activity Level:** Active
- Medical History & Pharmacotherapy:**
 - Underlying Secondary Medical Conditions:** Not Reported
 - History of Prior Low-Trauma Fractures:** No
 - Prescribed High-Risk Medications:** Not Reported

Clinical Risk Assessment Output:

Predicted Osteoporosis Probability: 13.2%

Low Risk (Score: 13.2%)

The patient is currently below the critical algorithmic threshold for osteoporosis.

Clinical Recommendations:

- Maintain routine age-appropriate wellness checks and balanced diet.

Footer: Disclaimer: OsteoGuard AI is an adjunctive decision support tool. All clinical decisions must be guided by verified BMD testing, FRAX scoring, and board-certified physician discretion.

8. Conclusion

The tuned Gradient Boosting pipeline provides strong predictive accuracy (87.76%) and high positive precision (98.05%), minimizing unnecessary clinical referrals while reliably flagging true positive cases. Encapsulating preprocessing and classification inside a single pipeline prevent data leakage and ensures seamless deployment in the Streamlit application.